Data Science & AI

DATA STRUCTURES Through Python

Practice Session

Lecture No.- 02

By- Satya sir

Topics to be Covered

– Practice Session

Practice Session – 2

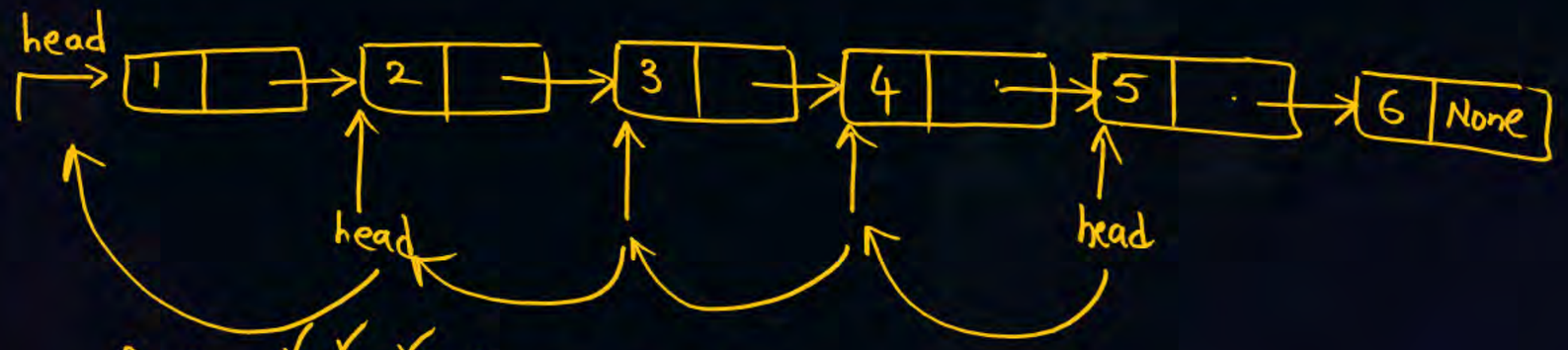
#Q. What does the following function print for a given Linked List with input 1,2,3,4,5,6?

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

def fun1(head):
    if head.next.next is None:
        return
    print(head.next.data, end=' ')
    fun1(head.next)
    print(head.data, end=' ')
```

fun(1, 2, 3, 4, 5, 6)
↳ fun(2, 3, 4, 5, 6)
↳ fun(3, 4, 5, 6)
↳ fun(4, 5, 6)
↳ fun(5, 6)

a) 2, 3, 4, 5, 6, 6, 5, 4, 3, 2
b) 2, 3, 4, 5, 5, 4, 3, 2
c) 2, 3, 4, 5, 4, 3, 2, 1
d) 3, 4, 5, 6, 6, 5, 4, 3



Print 2 3 4 5 4 3 2 1

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a) 3 8 5 13 11 10
b) 3 5 8 10 11 13
c) 3 8 16 13 24 50 ✓
d) 3 16 8 50 24 13

#Q. Consider the function foo and the binary tree shown.

```
class Node:
    def __init__(self, val=0, left=None, right=None):
        self.val = val
        self.left = left
        self.right = right

def foo(p):
    if p is None:
        return 0
    else:
        retval = p.val + foo(p.left) + foo(p.right)
        print(retval, end=" ")
        return retval
```

10 + 16 + 24 = 50

r = 10 + foo(L) + foo(r)

r = 16

r = 5 + f(L) + f(R)

r = 3 + f(L) + f(R)

0 0

3

r = 8 + f(L) + f(R)

8 + 0 + 0

r = 8

8

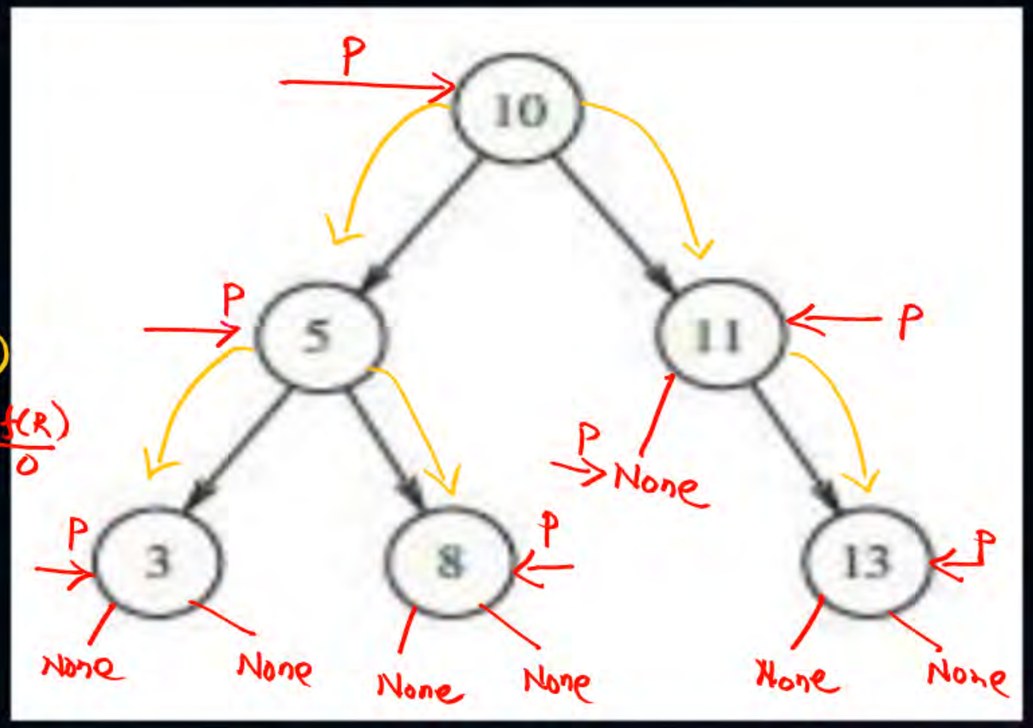
r = 11 + f(L) + f(R)

11 + 13 = 24

r = 0 + 13 + f(L) + f(R)

0 0

r = 13



o/p: 3, 8, 16, 13, 24, 50

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When foo is called with a pointer to the root node of the given binary tree, what will it print?

A. 3 8 5 13 11 10
B. 3 5 8 10 11 13
C. 3 8 16 13 24 50
D. 3 16 8 50 24 13

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#Q. Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in a non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is

array

~~2~~ 1	~~5~~ 2	~~3~~ 2	~~1~~ 3	4	~~2~~ 5	6

Resultant array
Expected

1	2	2	3	4	5	6

minimum distance = 3

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#Q. Which one of the following sequences when stored in an array at locations A[1], . . . , A[10] forms a max-heap?

A. 23, 17, 10, 6, 13, 14, 1, 5, 7, 12
B. 23, 17, 14, 7, 13, 10, 1, 5, 6, 12
C. 23, 17, 14, 6, 13, 10, 1, 5, 7, 15
D. 23, 14, 17, 1, 10, 13, 16, 12, 7, 5

A)

10 < 14
6 < 7
Not Max-heap

B)

Max-heap

C)

6 < 7

D)

1 < 7

Practice Session – 2- GATE 2024

#Q. You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T, and you do not have access to T.

Which one of the following statements is TRUE?

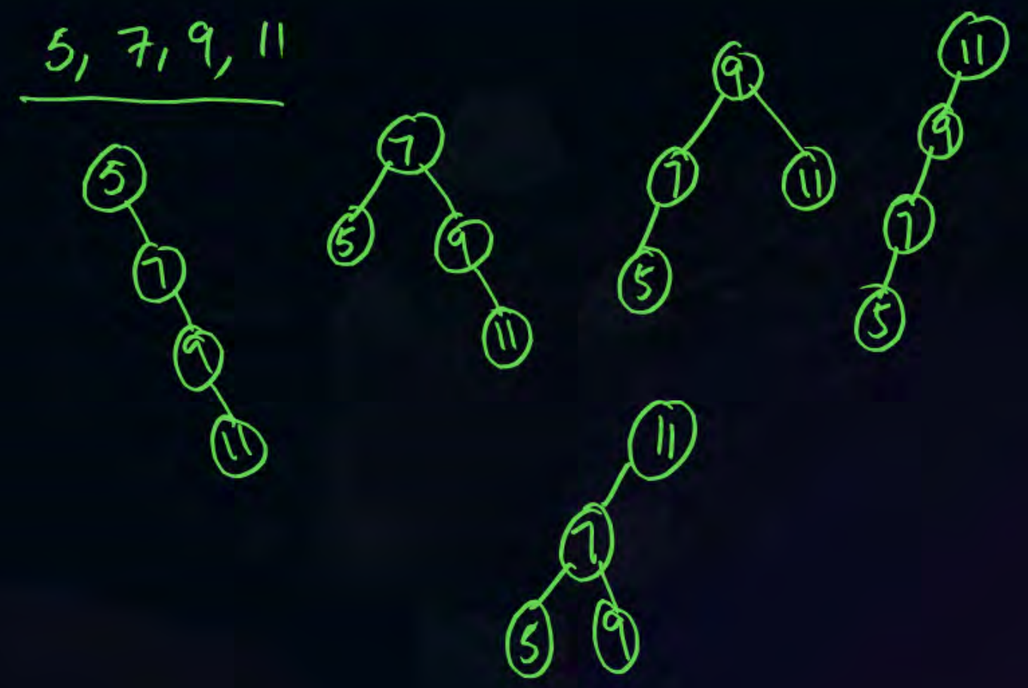
(A) In order traversal of T can be determined from V

(B) The root node of T can be determined from V

(C) Preorder traversal of T can be determined from V

(D) Post order traversal of T can be determined from V

5, 7, 9, 11



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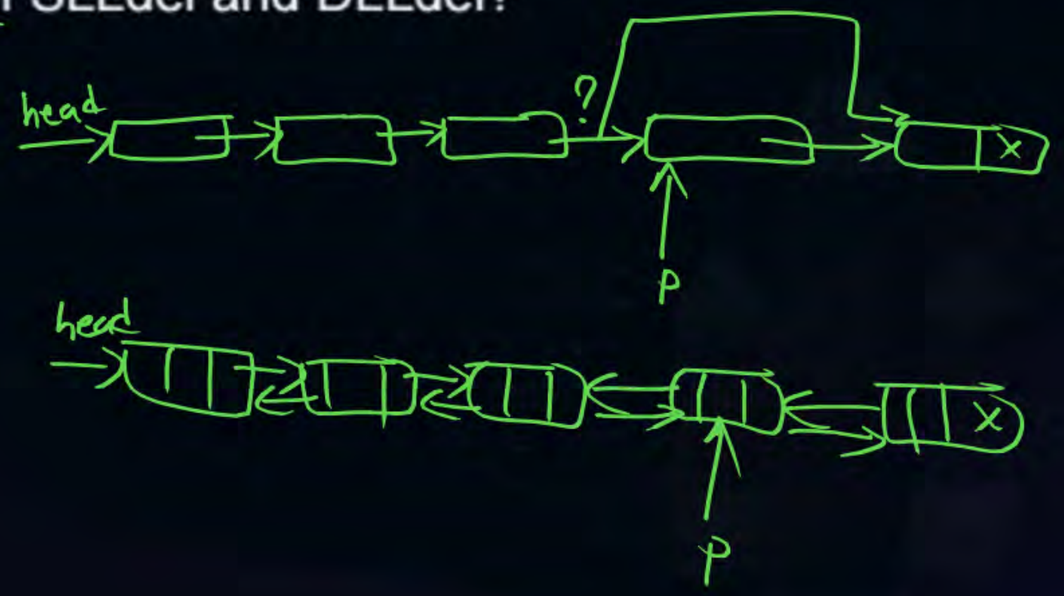
#Q. Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list.

Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?

A. SLLdel is O(1) and DLLdel is O(n)
B. Both SLLdel and DLLdel are O(log(n))
C. Both SLLdel and DLLdel are O(1)
D. SLLdel is O(n) and DLLdel is O(1)



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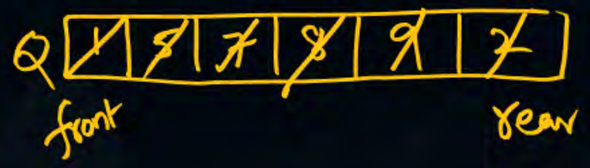
#Q. Consider a sequence a of elements a0 = 1, a1 = 5, a2 = 7, a3 = 8, a4 = ~~8~~ 9, and a5 = 2. The following operations are performed on a stack S and a queue Q, both of which are initially empty.

I: push the elements of a from a0 to a5 in that order into S.
II: enqueue the elements of a from a0 to a5 in that order into Q.
III: pop an element from S.
IV: dequeue an element from Q.
V: pop an element from S.
VI: dequeue an element from Q.
VII: dequeue an element from Q and push the same element into S.
VIII: Repeat operation VII three times.
IX: pop an element from S.
X: pop an element from S.



The top element of S after executing the above operations is 8.

Practice Session – 2

#Q. Convert the following infix expression to postfix expression

B * C - C + D / A / (E + E)

A. B C * C - D A / E E + / +
B. B C C - * D / A E E + / +
C. B C C - * D A / E E + / +
D. B C C * - D A / E E + / +

*	−	+	/	/	(	+	)
Pop	Pop	Pop	Pop	Pop			

Y: BC*C-DA/EE+/+

Practice Session – 2

#Q. Consider a min heap with elements 15, 30, 25, 40, 42, 27. After inserting elements 20 and 7, The Level order traversal sequence would be _____.

A. 7, 15, 20, 30, 40, 42, 27, 25
B. 7, 15, 20, 30, 42, 40, 27, 25
C. 7, 15, 20, 30, 27, 42, 40, 25
D. 7, 15, 20, 30, 42, 27, 25, 40

7, 15, 20, 30, 42, 27, 25, 40

Practice Session – 2

#Q. Consider a Max Heap with elements 90, 80, 70, 60, 50, 40. After 2 delete Operations, the level order traversal sequence would be _____

A. 70, 60, 50, 40
B. 70, 60, 40, 50
C. 70, 50, 60, 40
D. 70, 50, 40, 60

70, 60, 50, 40

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#Q. Consider the following tree traversals on a full binary tree: [MSQ]

(i) Preorder
(ii) Inorder
(iii) Postorder

Which of the following traversal options is/are sufficient to uniquely reconstruct the full binary tree?

(A) (i) and (ii)
(B) (ii) and (iii)
(C) (i) and (iii)
(D) (ii) only

A

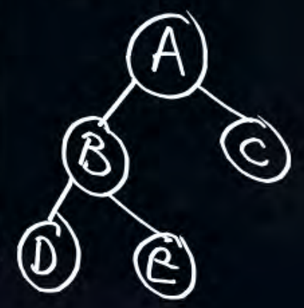
FBT

Pre order: A B C

Post order: B C A

Pre order: A B D E C

Post order: D E B C A



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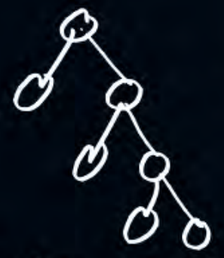

H=3
I=3
L=4
N=7

#Q. Let H,I,L, and N represent height, number of internal nodes, number of leaf nodes, and the total number of nodes respectively in a rooted binary tree.

Which of the following statements is/are always TRUE?

(A) $L \le I+1$ ✓
(B) $H+1 \le N \le 2^{H+1}-1$ ✓
(C) $H \le I \le 2^{H}-1$ $4 \le 7 \le 7$
(D) $H \le L \le 2^{H}-1$

$4 \le 15 \le 2^{3+1}-1$

$4 \le 15 \le 15$

H=3
I=7
L=8
N=15

h=3
h=2
h=1
h=0

L=1
H=4
$4 \le 1$ False.

H=4
I=4

Ans: A,B,C

THANK - YOU